

Incident definitions and thresholds

Generated from the committed trips.geojson (192,262 features; 10 crash points; 3,726 flagged braking segments).

Source: trips.geojson
Scope: Existing classified incidents and stored trip measurements

1. What this report is based on

This report does not rerun incident detection. Crash points are taken from GeoJSON features with **event_type = crash**. Sudden-braking rows are taken from segment features with **is_braking = true**. The report only derives contextual fields such as surrounding speed from timestamps and stored segment values.

2. Crash / fall definition

Rule / field	Current definition	Why it matters
Impact threshold	acc_y ≥ 6.0 g	Raw accelerometer impact spike enters crash-candidate clustering.
Impact clustering	Consecutive threshold samples grouped when gap ≤ 1.0 s	Prevents one physical impact from becoming many candidate spikes.
Burst grouping	Separate throws grouped when gap ≤ 3.0 s	Allows rapid multiple impacts to be evaluated as one burst.
Accelerometer settling	Post-impact window up to 3.0 s; stable for about 1.0 s; range ≤ 3.0 g	Requires the accelerometer signal to settle after the impact.
GPS stop	GPS speed ≤ 5.5 km/h for ≥ 1.0 s and coordinate spread ≤ 5.0 m	Separates a fall/stop from a jolt while continuing to ride.
Wheel motion before impact	Wheel must be turning before the burst when wheel diameter is available	Rejects impacts to a genuinely stationary bike.
Wheel stall / recovery	Wheel stall gap must be ≥ 1.0 s; recovery must be ≤ 120.0 s	Filters out events that look like parking/locking rather than a fall.
Impact-speed estimate	Wheel-rotation estimate; events at/above 40 km/h are not trusted	This is not the GNSS speed field.

3. Crash intensity / severity

Peak [g]	Label
6.0–<8.0 g	Minor
8.0–<11.0 g	Hard
≥11.0 g	Severe

4. Crash classification

Speed at / before impact	classification
speed_at_impact_unreliable = true	High-Speed Fall (speed estimate discarded, not measured)
≤1 km/h	Stationary Fall
>1–10 km/h	Low-Speed Fall
>10 km/h	High-Speed Fall
No trustworthy speed and no unreliable flag	Unclassified

The unreliable-flag check runs before any numeric-speed check, so an unreliable event is classified High-Speed Fall even though its speed_at_impact_kmh is null — that is why some High-Speed Fall rows in the crash table show no speed at impact. The frontend currently checks preimpact_speed_kmh first when present, then speed_at_impact_kmh. The committed generator currently exports speed_at_impact_kmh; therefore this report follows the same fallback logic without changing the event itself.

5. Crash outcome

outcome	Definition
Resolved	came_to_stop = true
Unresolved	unresolved = true
Unclassified	Neither outcome field establishes a state

6. Sudden braking definition

Rule / field	Current definition
Speed source	GNSS speed, smoothed with a 5-point rolling average
Braking intensity	(previous stored speed – current stored speed) / elapsed seconds; positive values only
Minimum segment time	0.5 s
GNSS braking threshold	≥ 2.0 km/h/s
Intensity cap	50 km/h/s
Implausible speed jump	> 20 km/h between consecutive stored speeds resets the braking series

The report uses the existing `is_braking` flag. It does not apply the separate 5 km/h/s hotspot threshold from the hotspot-generation workflow, because that threshold is for hotspot inclusion rather than the raw braking flag.

7. Derived report fields

Field	How this PDF obtains it
Surrounding speed	Mean of stored Speed values from 5s before to 1s after the incident timestamp (not a symmetric window — see <code>crash_row</code>).
Avg / peak pre-impact deceleration	Mean / maximum of stored <code>braking_intensity</code> values in the 5s before a crash. These are GNSS braking values, not accelerometer impact force.
Full crash timestamp	Matched from <code>crash_time_str</code> to the nearest segment timestamp on the same trip when a full timestamp is not already stored on the crash point.
Braking coordinates	First coordinate of the flagged <code>LineString</code> segment; no new spatial detection is performed.

8. Important limitations

- The crash point location is explicitly marked `location_approximate = true` in the exported data.
- Latitude/longitude are stored to 6 decimal places, but that is numerical storage precision, not GPS accuracy. The source code comments put the practical GPS accuracy floor at roughly 1–3 m.
- `speed_at_impact_kmh` is a wheel-rotation estimate rather than the smoothed GNSS Speed field.
- The GeoJSON does not contain a crash-event deceleration measurement. The crash-table deceleration columns are therefore explicitly pre-impact GNSS braking metrics.
- Vehicle type is not part of the current committed GeoJSON segment/crash schema, so it is not fabricated in these reports.
- The current generator drops a crash event outright when its wheel-rotation speed estimate reaches `CRASH_SPEED_CAP_KMH`, rather than keeping it with a null speed and a `speed_at_impact_unreliable` flag. Most crash points already committed in `trips.geojson` still carry that flag, so they were produced under an earlier version of this logic. Because already-processed trips are cached and skipped on later runs (see `load_existing_output` in `generate_trips_geojson.py`), these events will not be reprocessed under the current behavior unless `trips.geojson` is deleted and rebuilt from scratch.

9. Source files

- Executable definitions: `generate_trips_geojson.py`.
- Frontend classification/outcome logic: `app.js`.
- Data: `trips.geojson`.